What Each Step Adds: the Gnomons of the Golden Cycle and the Polar of the Keyed Pair



(working title — DICTO decides)

Theorem: one turn of DICTO's twelve-station golden cycle opens by doubling and closes by φ, and exactly four of its eleven steps add a golden multiple of the icosahedron; the keyed DICTONIC pair is the golden pair minus and plus the same eight Tricaps, and the golden pair's polars are cubes with face pyramids of heights 1/φ and 1/φ², which add up to Kepler's rhombic dodecahedron

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Paper #54 (proposed; DICTO decides) · Draft, Version 1 · 25 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/274868a/studies/2026-10-24-gnomon-cycle, …/studies/2026-10-23-shadow-forms/polar and …/papers/2026-10-25-gnomon-polar (research branch, commit 274868a)
The ideas are DICTO's: reading the golden cycle by what each step adds, from the gnomon method as described to DICTO by Júlia Pimentel, $(n + 1)^2 = n^2 + n + (n + 1)$; the hull-and-remainder polar of the keyed pair, "fill them in, maybe invert, then work out the remainder" (DICTO, 24 October 2026); and, from his earlier papers, the twelve-station golden cycle, the Cleo star, the DICTONIC SOLIDS and their names. Computation, exact proofs and drafting with Claude (Anthropic).

In short. Add one row and one column of dots to a square of dots, plus the corner dot, and you get the next square: 9 + 3 + 4 = 16. The L-shaped piece that is added is called a gnomon. DICTO's idea was to read his golden cycle of nested solids the same way, by looking only at what each step adds. The cycle has twelve stations, from an icosahedron out to the next icosahedron φ^3 times larger. We measured every station exactly. The first step adds exactly one more icosahedron, so it doubles; the last step multiplies by the golden ratio φ; and in between, two more steps add a clean golden amount, while seven do not. DICTO's second idea concerned his keyed pair of space-filling pieces, the Lotus Jewel Axes (with eight dents) and the Dodecaxes (with eight keys). A dented or keyed solid has no ordinary polar, so DICTO's method is to fill the dents in (or take the keys off), take the polar of the smooth solid, and keep what was removed as the remainder. The smooth solids are the golden pair of paper #48, and their polars are cubes with a low pyramid on every face. The two pyramid heights, $1/\varphi$ and $1/\varphi^2$, add up to exactly 1, the height that makes Kepler's rhombic dodecahedron; and the remainder is the same eight small Tricaps for both pieces. The two parts belong together: both read a solid by the piece that separates it from its neighbour.

Theorem A (gnomons of the golden cycle). In the cycle I ⊂ Cleo star ⊂ D′ ⊂ ID ⊂ OG ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ M ⊂ $I\cdot\varphi^3$ every volume is an exact number $a + b\sqrt{5}$ (a, b rational) times the volume of I. The cycle opens by doubling (Cleo star = 2·I) and closes by φ ($M = \varphi^8\cdot I$, so $I\cdot\varphi^3 = \varphi\cdot M$); $RT = 81\cdot I/\varphi^2$. Of the 66 volume ratios between stations exactly ten are a rational number times a power of φ, and of the eleven gnomons exactly four are a rational multiple of $\varphi^k\cdot I$: I, $(3/4)\varphi^{-2}\cdot I$, $18\varphi^{-4}\cdot I$ and $\varphi^7\cdot I$.

Theorem B (the keyed pair, hull and remainder). The Lotus Jewel Axes and the Dodecaxes are the Icosaxes and the Unborn Axes minus and plus the same eight Tricaps, each of volume $1/(12\varphi^2)$ [1]. The polar of the Icosaxes is a cube with six face pyramids of height 1/φ (in half cube edges), that of the Unborn Axes one with height $1/\varphi^2$; along the

whole factor-x family of [1] the two heights are x/(1 + x) and 1/(1 + x), so they always add up to 1, the rhombic dodecahedron's height. A polar of the Lotus Jewel Axes taken face by face about one centre would put spikes over the eight dents. All results are exact (55 checks, none failed).

Draft for DICTO. Number, title and the credit wording for Júlia Pimentel are DICTO's decisions. No published source for her method is recorded in the study; it is cited here as described to DICTO. Not published.

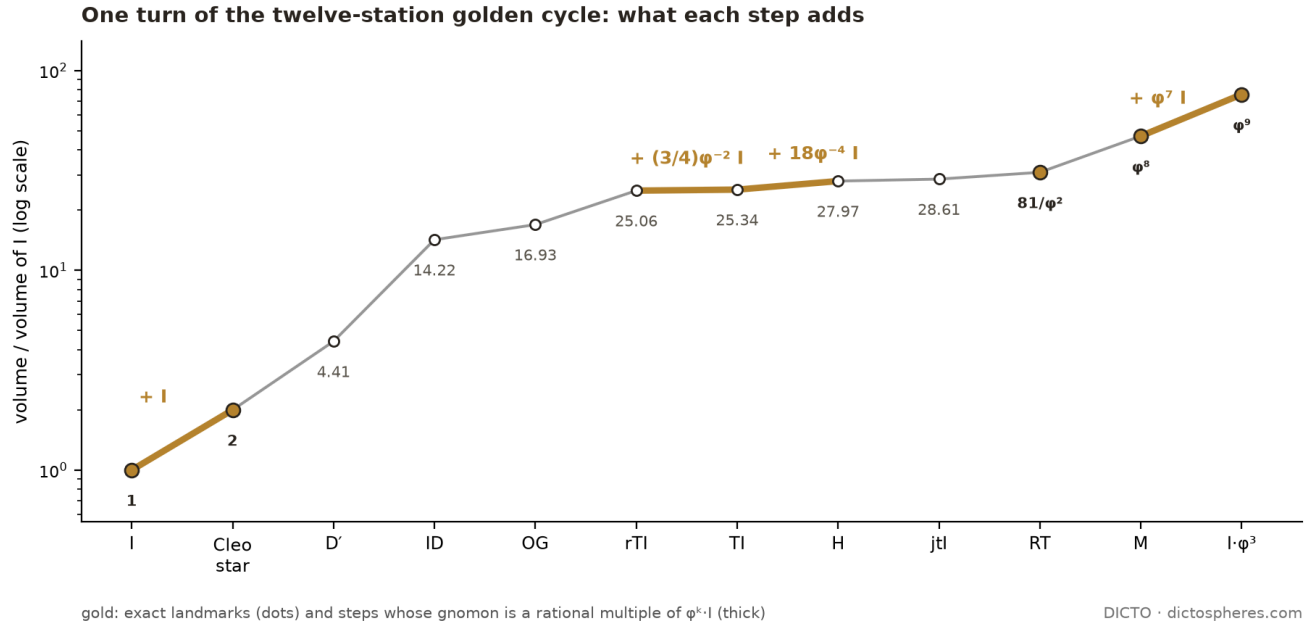


Figure 1. One turn of the twelve-station golden cycle (Table 1), each station as a multiple of the volume of the icosahedron I (log scale). Gold dots: the stations whose volume is a rational number times a power of φ times I. Thick gold steps: the four steps whose gnomon (what the step adds) is a rational multiple of $\varphi^k \cdot I$ (Theorem A).

1. Where it started, and why one paper

> *Both ideas came on 24 October 2026, and both look at the piece in between: the shell a step of the cycle adds, and the eight small pieces that a key or a dent moves from one solid to the other.*

The golden cycle grew over three of DICTO's papers. The Spheres of Dharma [2] nest classical icosahedral solids so that the next icosahedron, φ^3 times larger, closes one turn. Progenitors [3] set DICTO's Ogdoecontahedron (OG) and Hecatodihedron (H) [4] into the cycle, with the core and hull of their compound, the rectified truncated icosahedron (rTI) and its dual (jtI). Vajrayana [5] added DICTO's Cleo star and its hull D′, and closed the last loose step with the DICTO Metaxyhexecontahedron (M) [6], making twelve stations. On 24 October DICTO proposed reading the cycle by the gnomon method as described to him by Júlia Pimentel: in $(n + 1)^2 = n^2 + n + (n + 1)$ the next square is the last one plus what the step adds. Euclid uses the word gnomon for the L-shaped piece of a parallelogram [7, Book II, Definition 2], and growth by an added gnomon is the theme of D'Arcy Thompson's chapter on the equiangular spiral [8]. A float survey of the app's twelve stations found no single rule (step ratios from 1.01 to 3.23) and two clean ends; the exact study [S1] measured every station.

The same day DICTO turned to the keyed pair of paper #48 [1], the Lotus Jewel Axes and the Dodecaxes, and theorised a polar for them by his own method: fill the dents in, take the polar, and work out the remainder. The exact study is [S2]. We report the two together because they share one theme, the piece between two forms, and because each is short. In Part A the piece is the shell a step adds; in Part B it is the set of eight Tricaps between a keyed piece and its smooth golden form, and the polar heights of the two golden forms complement each other in the same way that the two keyed pieces do.

Part A. 2. The twelve stations, exactly

> *Every station is measured as a multiple of the innermost icosahedron. Ten of the twelve volumes come from the earlier papers; two, rTI and jtI, are measured here.*

The frame is that of [3]: the icosahedron I of the Spheres of Dharma has edge $1/\varphi$ and volume $(5/12)(\sqrt{5}-1)$. The volumes of I, the Cleo star, D′, ID, OG, TI, H, RT, M and $I\cdot\varphi^3$ are sums of the exact layer volumes published in [3, 5]. For rTI and jtI the study [S1] read the corners from the stored data of [3], recovered each coordinate exactly as $a + b\sqrt{5}$ with a, b in $(1/16)\mathbb{Z}$ (error below 10^{-6} in every case), computed jtI's other 32 corners exactly as the poles of TI's 32 face planes in the sphere $\rho^2 = 9/4$ [3], and summed exact signed tetrahedra over the convex hull. The same procedure reproduces the published volumes of TI, H, RT and $I\cdot\varphi^3$ exactly, at the same scale, and the float hull volumes agree.

Station	What it is	Volume (exact)	V / V(I), exact	≈
I	icosahedron of the Spheres of Dharma [2]	$-5/12 + (5/12)\sqrt{5}$	**1**	1
Cleo star	I with 20 Cleo-Icos as spikes [5]	$-5/6 + (5/6)\sqrt{5}$	**2**	2
D′	dodecahedron D at 2/3, hull of the Cleo star [5]	$10/9 + (14/27)\sqrt{5}$	$20/9 + (44/45)\sqrt{5}$	4.4086
ID	icosidodecahedron [2]	$175/48 + (79/48)\sqrt{5}$	$57/8 + (127/40)\sqrt{5}$	14.2245
OG	DICTO Ogdoecontahedron at $3\varphi^2/4$ [3]	$1975/456 + (895/456)\sqrt{5}$	$645/76 + (287/76)\sqrt{5}$	16.9309
rTI	rectified truncated icosahedron, core of the H–OG compound [3]	$-15/2 + (73/8)\sqrt{5}$	$183/8 + (39/40)\sqrt{5}$	25.0552
TI	truncated icosahedron [2]	$-35/4 + (39/4)\sqrt{5}$	$24 + (3/5)\sqrt{5}$	25.3416
H	DICTO Hecatodihedron [4]	$-365/4 + (189/4)\sqrt{5}$	$87 - (132/5)\sqrt{5}$	27.9678
jtI	dual of rTI, hull of the H–OG compound [3]	$-3465/38 + (900/19)\sqrt{5}$	$3321/38 - (999/38)\sqrt{5}$	28.6097
RT	rhombic triacontahedron [2]	$-135 + (135/2)\sqrt{5}$	$\mathbf{81/\varphi^2} = 243/2 - (81/2)\sqrt{5}$	30.9392
M	DICTO Metaxyhexecontahedron [5, 6]	$145/12 + (65/12)\sqrt{5}$	$\boldsymbol{\varphi^8} = 47/2 + (21/2)\sqrt{5}$	46.9787
$I\cdot\varphi^3$	the next icosahedron [2]	$235/12 + (35/4)\sqrt{5}$	$\boldsymbol{\varphi^9} = 38 + 17\sqrt{5}$	76.0132

Table 1. The twelve stations in the frame of [3] (I of edge $1/\varphi$). Bold: the volume ratios that are a rational number times a power of φ. The volumes grow strictly in this order.

3. Theorem A

> *The first step doubles, the last step multiplies by φ, and in between two more steps add a clean golden amount. Seven steps do not.*

> **Theorem A (gnomons of the golden cycle).** Let the stations and volumes be as in Table 1, and call $V_{i+1} - V_i$ the gnomon of step i.
> (i) **It opens by doubling:** Cleo star = 2·I; its gnomon is one I.

(ii) **It closes by φ:** $M = \varphi^8 \cdot I$, so $I \cdot \varphi^3 = \varphi^9 \cdot I = \varphi \cdot M$; the last gnomon is $\varphi^7 \cdot I$.
(iii) $RT = 81 \cdot I/\varphi^2$, hence $M/RT = \varphi^{10}/81$ and $I \cdot \varphi^3/RT = \varphi^{11}/81$.
(iv) Of the 66 ratios V_j/V_i ($i < j$) exactly ten are of the form $q \cdot \varphi^k$ with q rational and k an integer; all ten are among I, Cleo star, RT, M and $I \cdot \varphi^3$.
(v) Of the eleven gnomons exactly four are of the form $q \cdot \varphi^k \cdot I$: Cleo star − I = I, TI − rTI = $(3/4)\varphi^{-2} \cdot I$, H − TI = $18\varphi^{-4} \cdot I$ and $I \cdot \varphi^3 - M = \varphi^7 \cdot I$. The gnomons add up to $(\varphi^9 - 1) \cdot I$.

Proof. All volumes are in $\mathbb{Q}(\sqrt{5})$ (§2), and every identity is checked by exact arithmetic in $\mathbb{Q}(\sqrt{5})$ [S1, C]. (i) The spikes of the Cleo star are 20 Cleo-Icos, as many as make up I [5]. (ii) $\varphi^8 = (47 + 21\sqrt{5})/2$, and $M/I = 47/2 + (21/2)\sqrt{5}$ from Table 1; then $\varphi \cdot M = \varphi^9 \cdot I = (38 + 17\sqrt{5}) \cdot I$, which is $I \cdot \varphi^3$ because volume scales by $(\varphi^3)^3$. (iii) $RT/I \cdot \varphi^2 = 81$ with no $\sqrt{5}$ part. (iv)–(v) Write $x = a + b\sqrt{5}$. Since $\varphi^k = (L_k + F_k\sqrt{5})/2$ and $\varphi^{-k} = (-1)^k(L_k - F_k\sqrt{5})/2$, with F and L the Fibonacci and Lucas numbers, x is a rational multiple of a power of φ exactly when $b = 0$ ($k = 0$) or $b/a = \pm F_k/L_k$ for some $k \geq 1$. Because F_k and L_k have no common factor larger than 2, only the finitely many k with L_k at most twice the denominator of b/a need testing; the test is therefore exhaustive over all integers k, and it gives the lists stated. The gnomons telescope to $I \cdot \varphi^3 - I$. ∎

What the clean steps add. The first gnomon is the 20 spikes, one icosahedron. The step rTI → TI puts back the 60 corners that rectifying the truncated icosahedron cuts off (rTI's 90 corners lie on TI's 90 edges [3]): together $(3/4)\varphi^{-2} \cdot I$. The step TI → H adds the 20 Shields, the low hexagonal pyramids of H [4]: together $18\varphi^{-4} \cdot I$. The last step adds $\varphi^7 \cdot I$, the shell between M and the next icosahedron, filled by 60 pieces of one kind [5]. The seven other gnomons (Table 2) have no such form: so a gnomon rule in Júlia Pimentel's sense, one formula for every step, does not hold for this cycle; what holds is the frame, doubling at the start and φ at the end.

Step	Gnomon / I (exact)	≈	Form	Step ratio ≈
I → Cleo star	1	1.0000	**I**	2.0000
Cleo star → D′	2/9 + (44/45)√5	2.4086	—	2.2043
D′ → ID	353/72 + (791/360)√5	9.8159	—	3.2265
ID → OG	207/152 + (457/760)√5	2.7064	—	1.1903
OG → rTI	2187/152 − (2129/760)√5	8.1242	—	1.4798
rTI → TI	9/8 − (3/8)√5	0.2865	**$(3/4)\varphi^{-2} \cdot I$**	1.0114
TI → H	63 − 27√5	2.6262	**$18\varphi^{-4} \cdot I$**	1.1036
H → jtI	15/38 + (21/190)√5	0.6419	—	1.0230
jtI → RT	648/19 − (270/19)√5	2.3296	—	1.0814
RT → M	−98 + 51√5	16.0395	—	1.5184
M → $I \cdot \varphi^3$	29/2 + (13/2)√5	29.0344	**$\varphi^7 \cdot I$**	1.6180 = φ

Table 2. The eleven gnomons of one turn. The ten neat ratios of (iv): Cleo star/I = 2, $RT/I = 81/\varphi^2$, $M/I = \varphi^8$, $I \cdot \varphi^3/I = \varphi^9$, RT/Cleo star = $(81/2)/\varphi^2$, M/Cleo star = $\varphi^8/2$, $I \cdot \varphi^3$/Cleo star = $\varphi^9/2$, $M/RT = \varphi^{10}/81$, $I \cdot \varphi^3/RT = \varphi^{11}/81$, $I \cdot \varphi^3/M = \varphi$.

Part B. 4. The keyed pair and its golden forms

> *The keyed pieces are the smooth golden pieces with eight small corners moved from one to the other. Filling the dents in, or taking the keys off, gives back the smooth pair.*

Paper #48 [1] cut the gaps of a cubic lattice of icosahedra at the golden point and obtained the golden pair, the Icosaxes (the convex hull of all permutations of (±1/2, ±φ/2, 0), at the lattice points) and the Unborn Axes (all permutations of (±1/(2φ), ±φ/2, 0), at the cube centres (φ/2)(1, 1, 1) + φℤ³), both truncated octahedra; with the factor x of [1, Theorem 4] they are T(φ) and T(1/φ) of Koca's orbit [9]. In the Lotus Jewel network the same cut gives the keyed pair: the Lotus Jewel Axes, the Icosaxes with eight Tricaps removed (dents), and the Dodecaxes, the Unborn Axes with the same eight Tricaps added (keys) [1, Theorem 3]. The study [S2] rechecks this exactly and records where each Tricap sits: its three base corners are corners of the Unborn, its apex is a corner of the cube inscribed in the Unborn's dodecahedron, its base lies in a hexagon shared by the two pieces, and the eight dents lie inside the icosahedron, pairwise apart. The Tricap has volume $(3 - \sqrt{5})/24 = 1/(12\varphi^2)$, the eight together $1 - \sqrt{5}/3 \approx 0.2546$; so $\sqrt{5} = (1 + 2\sqrt{5}/3) - 8/(12\varphi^2)$ and $2 = (1 + \sqrt{5}/3) + 8/(12\varphi^2)$, and the sum is φ^3, one lattice cell.

A solid with dents has no ordinary polar. DICTO's method reads it through its smooth form: the Icosaxes is the convex hull of the Lotus Jewel Axes, the Unborn Axes is the Dodecaxes with its keys off, and the remainder is the eight Tricaps in both cases.

5. Theorem B

> *Turned inside out, each golden piece becomes a cube with a low pyramid on every face. The two pyramid heights always add up to the height that makes Kepler's rhombic dodecahedron.*

> **Theorem B (hull and remainder).** Take polars in the unit sphere about each piece's own centre, and measure pyramid heights in half cube edges.
> (i) For every $x > 0$ the polar of T(x) is a cube with a square pyramid of height $x/(1 + x)$ on each face, and the polar of its partner T(1/x) one with height $1/(1 + x)$. The two heights add up to 1, the height of the pyramids that turn a cube into Kepler's rhombic dodecahedron [10].
> (ii) For the golden pair: the polar of the Icosaxes has pyramids of height $1/\varphi$ and that of the Unborn Axes of height $1/\varphi^2$, and $1/\varphi + 1/\varphi^2 = 1$. Both polars have their apexes at distance $2/\varphi$ from the centre; their cube corners lie at $2\sqrt{3}/\varphi^2$ and $2\sqrt{3}/\sqrt{5}$.
> (iii) In DICTO's hull-and-remainder reading the Lotus Jewel Axes is the polar of (ii) for the Icosaxes with the eight Tricaps as remainder, and the Dodecaxes that of the Unborn Axes with the same eight Tricaps as remainder.
> (iv) Taken face by face about the one centre, the three walls of each dent of the Lotus Jewel Axes have their poles at distance $\sqrt{(10 - 2\sqrt{5})} = 4 \sin 36° \approx 2.351$, against $2\sqrt{3}/\varphi^2 \approx 1.323$ for the hexagon the dent sits in, and ≈ 37.38° away from its direction.

Proof. (i) Let T be the convex hull of all permutations of (±a, ±b, 0) with $0 < a < b$. Its six squares lie in the planes $\pm x_i = b$ and its eight hexagons in $\pm x \pm y \pm z = a + b$. Their poles are $(\pm 1/b)e_i$ and the cube corners $(\pm 1, \pm 1, \pm 1)/(a + b)$, so the cube has half edge $1/(a + b)$ and the pyramid apexes stand at $1/b$; the height in half edges is $(a + b)/b - 1 = a/b$. For T(x), (a, b) = (x, 1 + x) up to scale, which gives $x/(1 + x)$; for T(1/x), (1/x, 1 + 1/x) gives $1/(1 + x)$. The sum is 1 for every x. (ii) The Icosaxes has $a/b = (1/2)/(\varphi/2) = 1/\varphi$ and the Unborn Axes $(1/(2\varphi))/(\varphi/2) = 1/\varphi^2$; the distances follow from 1/b and $\sqrt{3}/(a + b)$. (iii) is the definition of the reading together with [1, Theorem 3] as rechecked in [S2]. (iv) The dent facing the cube centre $(\varphi/2)(1, 1, 1)$ has apex $(1, 1, 1)/(2\varphi)$ and base corners $(0, 1/2, \varphi/2)$, $(\varphi/2, 0, 1/2)$, (1/2,

φ/2, 0), which are Icosaxes corners on the hexagon x + y + z = φ²/2; the pole n/(n·A) of each wall plane n·X = n·A has squared length 10 − 2√5, computed exactly [C]; the angle is computed from these exact vectors. ∎

Remarks. At Kelvin's cell, x = 1, both heights are 1/2: the polar is the tetrakis hexahedron, the classical dual of the truncated octahedron [11]; the golden pair moves the two heights apart, to 1/φ and 1/φ², without changing their sum. Part (iv) is why the reading of (iii) is used: a one-centre polar of the dented solid would carry three-pointed spikes at all eight dents, far outside the cube corners. DICTO doubted that spikes belong to the polar, and the hull-and-remainder reading has none. The two golden polars do not fill space together: their edge angles, ≈ 136.35° and 153.43° for the Icosaxes' polar and ≈ 150.77° and 131.81° for the Unborn Axes' polar, allow no combination round an edge that makes 360° (float count, [S2], polarfill1.py). The study also computed the polar of one Tricap about its own centroid, a three-sided pyramid with edges in the ratio 1 : tan 72°; since that centre is a free choice, it is recorded only as a side fact.

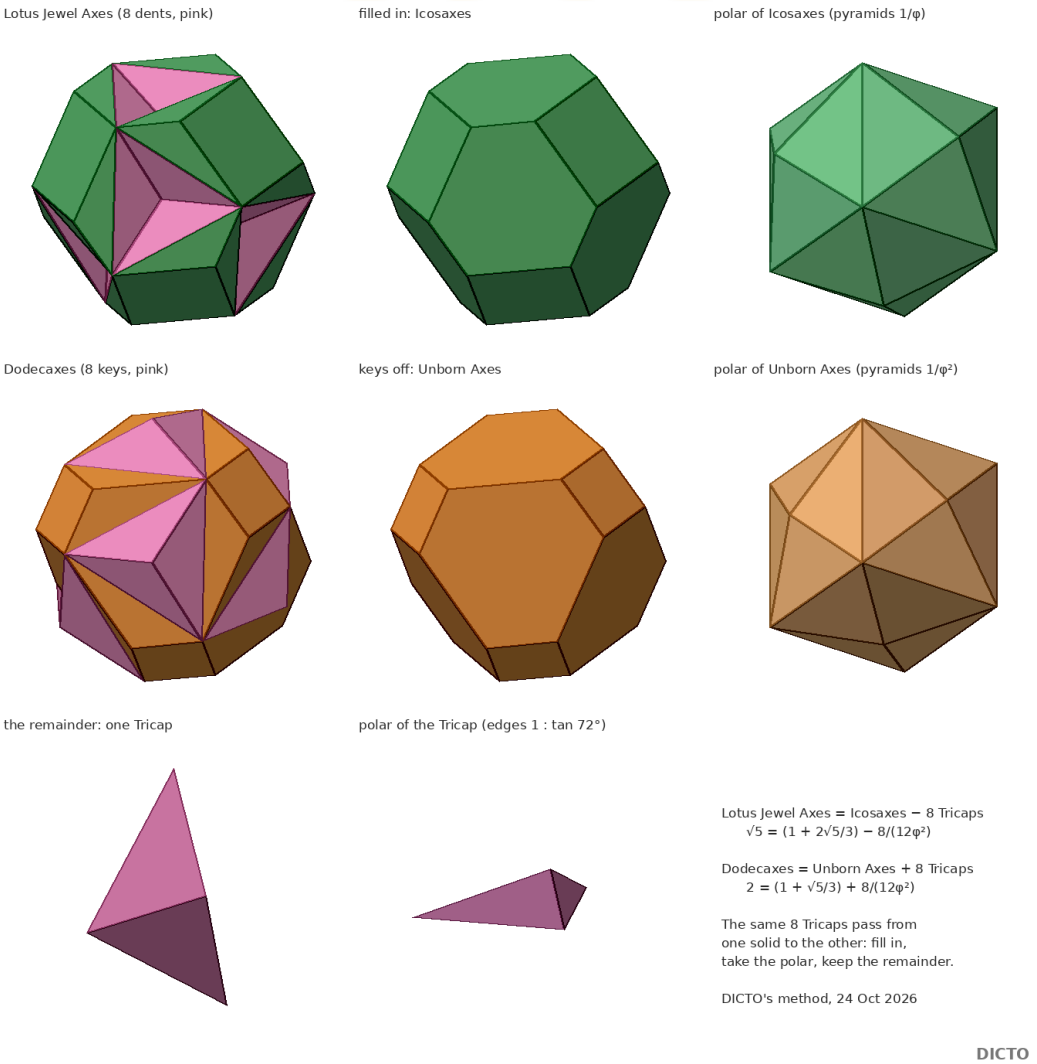


Figure 2. DICTO's hull-and-remainder polar [S2]. Top row: the Lotus Jewel Axes with its 8 dents (pink), filled in to the Icosaxes, and the Icosaxes' polar (pyramids 1/φ). Middle row: the Dodecaxes with its 8 keys (pink), keys off to the Unborn Axes, and its polar (pyramids 1/φ²). Bottom: the remainder, one of the eight Tricaps, and its polar about its own centroid (side fact only, see Remarks). Exact proof: keyed_polar_exact.py, 15 checks (Tricaps drawn in the orientation of [1]).

6. Conclusion

> *Read by what lies between, the golden cycle shows a clean beginning and end and an irregular middle, and the keyed pair shows one remainder shared by two pieces whose polars complement each other.*

DICTO's gnomon reading of the twelve-station cycle gives an exact frame: one turn opens by doubling and closes by φ, RT sits at $81/\varphi^2$ and M at φ^8, and only four of the eleven shells (the spikes, the 60 corners of TI, the 20 Shields and the last shell) add a golden multiple of the icosahedron. DICTO's hull-and-remainder polar reads the keyed pair through the golden pair: the polars are cubes with pyramids of heights $1/\varphi$ and $1/\varphi^2$, which add up to Kepler's rhombic dodecahedron, as every pair of the factor-x family does, and the eight Tricaps are the common remainder. Open points: whether some other choice of stations, or of what counts as a step, gives a gnomon rule through the middle of the cycle; and whether pyramid-capped cubes with heights adding to 1, at other sizes or with the Tricaps as separate pieces, can fill space together.

7. What is known, and what is not found

> *The solids, the gnomon and polar duality are classical. What we did not find is this reading of DICTO's cycle and pair.*

Known. The gnomon as the piece added to a figure [7], and growth by gnomons [8]; the volumes of I, ID, TI and RT, and the rectified truncated icosahedron and its dual as solids (noted in [3]); the station volumes of I, Cleo star, D′, ID, OG, TI, H, RT, M and $I\cdot\varphi^3$ [2–6], and Cleo star = 2·I [5]; polar reciprocity of convex polyhedra and the tetrakis hexahedron as the dual of the truncated octahedron [11]; the rhombic dodecahedron as a cube with six pyramids of height half its edge [10]; the T(x) orbit [9]; the golden and keyed pairs, the Tricap and its volume [1].

Not found, in the sources searched (below): the exact volumes of rTI and jtI as stations of this cycle; $M = \varphi^8\cdot I$ and $RT = 81\cdot I/\varphi^2$; the gnomon table and its four golden gnomons; the polar heights $1/\varphi$ and $1/\varphi^2$ of the golden pair and the sum-to-one along the factor-x family. Part B (i) is elementary and may well be known in some form; "not found" is limited to these sources and is not a claim of novelty.

Sources searched, 25 October 2026. DICTO's reference library (ask-library, eight queries: gnomon and figurate numbers; golden volume ratios of nested icosahedral solids; truncated octahedron and tetrakis hexahedron as polars; polar reciprocation; rectified truncated icosahedron volume; polars of non-convex solids; Euclid Book II and Heath; Aristotle on the gnomon), which holds Heath's Euclid volume III but not volume I, and no gnomon or polar-height source; DICTO's own papers [1–6] and Collected Geometric Works, Version 24; Crossref for every DOI below. No web search was made for this draft.

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[S1] Study "Gnomon rule, golden cycle", studies/2026-10-24-gnomon-cycle (gnomon_exact.py, gnomon_table.py, gnomon_all12.py; commits afd8a42, 4b05e6f, 4cac277).
[S2] Study "Shadow forms", studies/2026-10-23-shadow-forms/polar (keyed_polar_exact.py, keyed_polar1.py, keyed_polar2.py, polarfill1.py; commits 4c18a0b, e9aed63).
[C] This paper's own check script, papers/2026-10-25-gnomon-polar/checks.py → checks.txt.

Appendix. Checks

Script	What it checks	PASS	FAIL
gnomon_exact.py [S1]	star = 2I, I·φ³ = φ·M, M = φ⁸I, I·φ³ = φ⁹I	4	0
gnomon_all12.py [S1]	method reproduces TI, H, RT, I·φ³; rTI and jtI exact vs float hulls; jtI = hull(TI ∪ PD), 32 poles; strict growth	9	0
gnomon_table.py [S1]	layer sums of [3] reproduce RT (assertion)	1	0
keyed_polar_exact.py [S2]	Tricap volume and position, dents and keys, √5 and 2, polar heights 1/φ and 1/φ², tan 72°	15	0
checks.py [C]	every number in this paper: Tables 1–2, the exhaustive φ-power test, Theorem B (i)–(iv) symbolically	26	0
total (exact)		**55**	**0**

All scripts re-run on dicto-big-brother on 25 October 2026 with output identical to the saved files (the saved gnomon-table.txt carries one extra line, RT/I·φ² = 81, which is check RT = 81·I/φ² of [C]). Exact arithmetic: Python fractions over ℚ(√5) and sympy; floats only for the hull search (whose exact corner sets are then used), the angles of Theorem B (iv) and the Remarks, and the figures.